

Case ID: COL-2026-013

Patient ID: PT-2026-0847

Patient Name: Sample Patient

Tissue: Colorectal

Stain: H&E

Date: 31 Aug 2026 | Status: IN PROGRESS

Analyzed Specimen & Image Quality

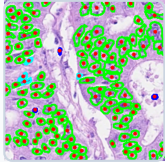


Image ID: IMG-001 | Source: 55c052d5-90ec-46ae-a5f3-34e8778a6fb5-1_all_104801

Dimensions: 1080 × 1068 px | Magnification: 40×

Quality: GOOD | Blur: Acceptable | Calibration: Passed

AI Model: HoVer-Net Instance Segmentation (Green=Boundaries, Red=Epithelial)

Quality Assessment: Accepted (Passed Diagnostic Quality Standard)

Quantitative Morphology Summary

Nuclei Count: 1824

Nuclear Density: 139.2 /mm²

Circularity: 0.72

Gland Count: 146

Mean Gland Area: 2840.0 px²

Boundary Irregularity: 0.64

Reference vs Patient Comparison Table

Metric	Reference Baseline	Patient Specimen
Nuclei Count	1200	1824
Nuclear Density	98.5 /mm ²	139.2 /mm ²
Mean Nuclear Area	38.6 px ²	47.3 px ²
Nuclear Circularity	0.86	0.72
Gland Count	162	146
Gland Density	12.8 /mm ²	11.2 /mm ²
Mean Gland Area	3,420 px ²	2,840 px ²
Gland Irregularity	0.31	0.64

AI-Assisted Interpretation

Structured computational observations suggest architectural changes consistent with adenomatous features. Deviations observed in critical morphology metrics.

Supporting Evidence

- High structural similarity to REF-021 (Adenoma-like morphology, 94.2%)
- Increased nuclear-to-cytoplasmic ratio observed in 48.9% of epithelial cells
- Significant gland crowding with moderate branching

PATHOLOGIST REVIEW REQUIRED

Morphological deviations exceed typical baseline parameters. Comprehensive manual review of all slides is strongly recommended.